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DBMS Lab Assignment - 2

Title: Introduction to DDL and DML commands and its execution.

Description:

Data definition language defines the schema for the database by specifying entities and the relationship among them. In addition to this, DDL even defines certain security constraints.The execution of DDL statements results in new tables which are stored in "system catalog" also called data dictionary or data directory.

Data Manipulation Language is a language that provides a set of operations to support the basic data manipulation operations on the data held in the databases. It allows users to insert, update, delete and retrieve data from the database. Data manipulations are applied at internal, conceptual and external levels of schemas. However, the level of complexity at each schema level varies from one another.

Data Control Language statements control access to data and the database using statements such as GRANT and REVOKE. A privilege can either be granted to a User with the help of GRANT statement. The privileges assigned can be SELECT, ALTER, DELETE, EXECUTE, INSERT, INDEX etc. In addition to granting of privileges, you can also revoke (taken back) it by using REVOKE command.

DDL : Data Definition Language

All DDL commands are auto-committed. That means it saves all the changes permanently in the database.

Command	Description
create	to create new table or database
alter	for alteration
truncate	delete data from table
drop	to drop a table
rename	to rename a table

DML : Data Manipulation Language

DML commands are not auto-committed. It means changes are not permanent to database, they can be rolled back.

Command	Description
insert	to insert a new row
update	to update existing row
delete	to delete a row
merge	merging two rows or two tables

DCL : Data Control Language

Data control language provides command to grant and take back authority.

Command	Description
grant	grant permission of right
revoke	take back permission.

Assignment 2 Lab Questions

Section- 1 DDL Commands

Q-1 Explain DDL commands and their syntax

Data Definition Language (DDL) commands are used to define, modify, and delete the structure of database objects (such as tables, indexes, and views). They deal with the schema rather than the actual data.

1. CREATE

- **Definition:** Used to build a new table or database object.
- **Syntax:**

SQL

```
CREATE TABLE table_name (
    column1 datatype,
    column2 datatype
);
```

- **Example:**

SQL

```
CREATE TABLE Student (
    id INT,
    name VARCHAR(50)
);
```

2. ALTER

- **Definition:** Used to modify the structure of an existing table (like adding a column).
- **Syntax:**

SQL

```
ALTER TABLE table_name ADD column_name datatype;
```

- **Example:**

SQL

```
ALTER TABLE Student ADD age INT;
```

3. TRUNCATE

- **Definition:** Deletes **all data rows** inside a table, but keeps the table structure intact.
- **Syntax:**

  SQL

  ```
  TRUNCATE TABLE table_name;
  ```

- **Example:**

  SQL

  ```
  TRUNCATE TABLE Student;
  ```

4. DROP

- **Definition:** Completely deletes a table along with all its data and structure permanently.
- **Syntax:**

  SQL

  ```
  DROP TABLE table_name;
  ```

- **Example:**

  SQL

  ```
  DROP TABLE Student;
  ```

5. RENAME

- **Definition:** Changes the name of an existing table or column.
- **Syntax:**

  SQL

  ```
  RENAME TABLE old_table_name TO new_table_name;
  ```

- **Example:**

  SQL

  ```
  RENAME TABLE Student TO Learner;
  ```

Q-2 Create tables for the following relational model of library management. Apply constraints on the columns and also alter the structure according to your requirements.

1. **SIULIBRARY** (<u>Slid</u>,lname,location,noofbranches)

2. **Ilibrary**(Lid, lname, city, area, slid)
3. **BOOKS**(Bid, Bname, Price , Lid)
4. **Noofcopies**(bnid,bid,blid)
5. **AUTHOR**(Aid, Aname,email,phoneno)
6. **Writes**(Bid, Aid, pid)
7. **PUBLISHER**(Pid, Pname)
8. **SELLER**(Sid, slname, city)
9. **DEPARTMENT** (Deptid,deptname, Iname,lid)
10. **STUDENT**(Stuid, Sname, email,memid deptid)
11. **STAFF**(Stid, stname, email, deptid,memid)
12. **PURCHASE**(prid , lid, sid, pid, bid, quantity ,date, totalcost)
13. **ISSUE**(lssueid, memid, bid, lid, issuedate, returndate)
14. **SELLS** (sid, bid,pid)
15. **Employee**(eid,empname,email,salary,lid)
16. **A_specialization**(spec_id,spec_name,Aid)
17. **Member**(em

id,lid)

Section- 2 DML Command execution

Q-1. Explain DML commands and their syntax.

Data Manipulation Language (DML) commands are used to modify, retrieve, and manage data stored within existing database tables. Unlike DDL (which works with the database structure), DML deals directly with the actual data rows.

1. INSERT

- **Definition:** Used to insert new records or rows of data into a table.
- **Syntax:**

SQL

```
INSERT INTO table_name (column1, column2, ...)
VALUES (value1, value2, ...);
```

- **Example:**

SQL

```
INSERT INTO STUDENT (Stuid, Sname, email)
VALUES (101, 'Aarav Sharma', 'aarav@gmail.com');
```

2. UPDATE

- **Definition:** Used to modify existing data in a table based on a specific condition.
- **Syntax:**

SQL

```
UPDATE table_name
SET column1 = value1, column2 = value2, ...
WHERE condition;
```

- **Example:**

SQL

```
UPDATE STUDENT
SET email = 'aarav_new@gmail.com'
WHERE Stuid = 101;
```

3. DELETE

- **Definition:** Used to remove one or more existing records from a table based on a condition.
- **Syntax:**

SQL

```
DELETE FROM table_name
WHERE condition;
```

- **Example:**

SQL

```
DELETE FROM STUDENT
WHERE Stuid = 101;
```

4. MERGE

- **Definition:** Used to perform insert, update, or delete operations on a target table conditionally based on matching results from a source table (often referred to as an "upsert").
- **Syntax:**

SQL

```
MERGE INTO target_table USING source_table
ON (condition)
WHEN MATCHED THEN
    UPDATE SET target_table.column1 = source_table.column1
WHEN NOT MATCHED THEN
    INSERT (column1, column2) VALUES (source_table.column1,
source_table.column2);
```

- **Example:**

SQL

```
MERGE INTO STUDENT target
USING New_Student_Data source
ON (target.Stuid = source.Stuid)
WHEN MATCHED THEN
    UPDATE SET target.email = source.email
WHEN NOT MATCHED THEN
    INSERT (Stuid, Sname, email)
    VALUES (source.Stuid, source.Sname, source.email);
```

Q-2. Insert 5 tuples in each of the created tables.

Q-3 Execute following queries on the library database

1. Which institute libraries are located in pune city?
2. To which institute CS department belongs to?
3. Find all the books whose price is between 800 to 12000?
4. Find out such employees who’s salaries are not greater than 50,000/-
5. Find out such sellers who’s name end with “ta”
6. Find out such institute libraries where their area information is missing.
7. Find out such staff members who’s name doesn’t starts with “A”
8. Find out such SIU libraries which have institute libraries located in Bangalore.
9. Which students belong to civil department?
10. Find out books which are written by “shruti” and published by Mcgraw hill ?

Input and Output:

Section-1 (Q-2)

```
CREATE TABLE SIULIBRARY (
    Slid INT PRIMARY KEY,
    lname VARCHAR(100) NOT NULL,
    location VARCHAR(100),
    noofbranches INT CHECK (noofbranches >= 0)
);

CREATE TABLE Ilibrary (
    Lid INT PRIMARY KEY,
    lname VARCHAR(100) NOT NULL,
    city VARCHAR(50),
    area VARCHAR(50),
    slid INT,
    CONSTRAINT fk_ilibrary_siulibrary FOREIGN KEY (slid) REFERENCES SIULIBRARY(Slid)
ON DELETE SET NULL
);

CREATE TABLE BOOKS (
    Bid INT PRIMARY KEY,
    Bname VARCHAR(150) NOT NULL,
    Price DECIMAL(10, 2) CHECK (Price >= 0),
    Lid INT,
    CONSTRAINT fk_books_ilibrary FOREIGN KEY (Lid) REFERENCES Ilibrary(Lid) ON
DELETE CASCADE
);

CREATE TABLE Noofcopies (
    bnid INT PRIMARY KEY,
    bid INT,
    blid INT,
    CONSTRAINT fk_noofcopies_books FOREIGN KEY (bid) REFERENCES BOOKS(Bid),
```

```sql
    CONSTRAINT fk_noofcopies_ilibrary FOREIGN KEY (blid) REFERENCES Ilibrary(Lid)
);

CREATE TABLE AUTHOR (
    Aid INT PRIMARY KEY,
    Aname VARCHAR(100) NOT NULL,
    email VARCHAR(100) UNIQUE,
    phoneno VARCHAR(15)
);

CREATE TABLE PUBLISHER (
    Pid INT PRIMARY KEY,
    Pname VARCHAR(100) NOT NULL
);

CREATE TABLE Writes (
    Bid INT,
    Aid INT,
    pid INT,
    PRIMARY KEY (Bid, Aid, pid),
    CONSTRAINT fk_writes_books FOREIGN KEY (Bid) REFERENCES BOOKS(Bid),
    CONSTRAINT fk_writes_author FOREIGN KEY (Aid) REFERENCES AUTHOR(Aid),
    CONSTRAINT fk_writes_publisher FOREIGN KEY (pid) REFERENCES PUBLISHER(Pid)
);

CREATE TABLE SELLER (
    Sid INT PRIMARY KEY,
    slname VARCHAR(100) NOT NULL,
    city VARCHAR(50)
);

CREATE TABLE DEPARTMENT (
    Deptid INT PRIMARY KEY,
    deptname VARCHAR(100) NOT NULL,
    Iname VARCHAR(100),
    lid INT,
```

```
    CONSTRAINT fk_dept_ilibrary FOREIGN KEY (lid) REFERENCES Ilibrary(Lid)
);

CREATE TABLE Member (
    memid INT PRIMARY KEY,
    lid INT,
    CONSTRAINT fk_member_ilibrary FOREIGN KEY (lid) REFERENCES Ilibrary(Lid)
);

CREATE TABLE STUDENT (
    Stuid INT PRIMARY KEY,
    Sname VARCHAR(100) NOT NULL,
    email VARCHAR(100) UNIQUE,
    memid INT,
    deptid INT,
    CONSTRAINT fk_student_member FOREIGN KEY (memid) REFERENCES Member(memid),
    CONSTRAINT fk_student_dept FOREIGN KEY (deptid) REFERENCES
DEPARTMENT(Deptid)
);

CREATE TABLE STAFF (
    Stid INT PRIMARY KEY,
    stname VARCHAR(100) NOT NULL,
    email VARCHAR(100) UNIQUE,
    deptid INT,
    memid INT,
    CONSTRAINT fk_staff_dept FOREIGN KEY (deptid) REFERENCES DEPARTMENT(Deptid),
    CONSTRAINT fk_staff_member FOREIGN KEY (memid) REFERENCES Member(memid)
);

CREATE TABLE PURCHASE (
    prid INT PRIMARY KEY,
    lid INT,
    sid INT,
    pid INT,
    bid INT,
```

```
    quantity INT CHECK (quantity > 0),
    date DATE,
    totalcost DECIMAL(12, 2),
    CONSTRAINT fk_purchase_ilibrary FOREIGN KEY (lid) REFERENCES Ilibrary(Lid),
    CONSTRAINT fk_purchase_seller FOREIGN KEY (sid) REFERENCES SELLER(Sid),
    CONSTRAINT fk_purchase_publisher FOREIGN KEY (pid) REFERENCES PUBLISHER(Pid),
    CONSTRAINT fk_purchase_books FOREIGN KEY (bid) REFERENCES BOOKS(Bid)
);

CREATE TABLE ISSUE (
    lssueid INT PRIMARY KEY,
    memid INT,
    bid INT,
    lid INT,
    issuedate DATE,
    returndate DATE,
    CONSTRAINT fk_issue_member FOREIGN KEY (memid) REFERENCES Member(memid),
    CONSTRAINT fk_issue_books FOREIGN KEY (bid) REFERENCES BOOKS(Bid),
    CONSTRAINT fk_issue_ilibrary FOREIGN KEY (lid) REFERENCES Ilibrary(Lid)
);

CREATE TABLE SELLS (
    sid INT,
    bid INT,
    pid INT,
    PRIMARY KEY (sid, bid, pid),
    CONSTRAINT fk_sells_seller FOREIGN KEY (sid) REFERENCES SELLER(Sid),
    CONSTRAINT fk_sells_books FOREIGN KEY (bid) REFERENCES BOOKS(Bid),
    CONSTRAINT fk_sells_publisher FOREIGN KEY (pid) REFERENCES PUBLISHER(Pid)
);

CREATE TABLE Employee (
    eid INT PRIMARY KEY,
    empname VARCHAR(100) NOT NULL,
    email VARCHAR(100) UNIQUE,
    salary DECIMAL(10, 2) CHECK (salary >= 0),
```

```
    lid INT,
    CONSTRAINT fk_emp_ilibrary FOREIGN KEY (lid) REFERENCES Ilibrary(Lid)
);

CREATE TABLE A_specialization (
    spec_id INT PRIMARY KEY,
    spec_name VARCHAR(100) NOT NULL,
    Aid INT,
    CONSTRAINT fk_aspec_author FOREIGN KEY (Aid) REFERENCES AUTHOR(Aid) ON
DELETE CASCADE
);
```

Command Prompt - mysql -u

```
mysql> use LibraryDB;
Database changed
mysql> CREATE TABLE SIULIBRARY (
    ->     Slid INT PRIMARY KEY,
    ->     lname VARCHAR(100) NOT NULL,
    ->     location VARCHAR(100),
    ->     noofbranches INT CHECK (noofbranches >= 0)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> CREATE TABLE Ilibrary (
    ->     Lid INT PRIMARY KEY,
    ->     lname VARCHAR(100) NOT NULL,
    ->     city VARCHAR(50),
    ->     area VARCHAR(50),
    ->     slid INT,
    ->     CONSTRAINT fk_ilibrary_siulibrary FOREIGN KEY (slid) REFERENCES SIULIBRARY(Slid) ON DELETE SET NULL
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> CREATE TABLE BOOKS (
    ->     Bid INT PRIMARY KEY,
    ->     Bname VARCHAR(150) NOT NULL,
    ->     Price DECIMAL(10, 2) CHECK (Price >= 0),
    ->     Lid INT,
    ->     CONSTRAINT fk_books_ilibrary FOREIGN KEY (Lid) REFERENCES Ilibrary(Lid) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> CREATE TABLE Noofcopies (
    ->     bnid INT PRIMARY KEY,
    ->     bid INT,
    ->     blid INT,
    ->     CONSTRAINT fk_noofcopies_books FOREIGN KEY (bid) REFERENCES BOOKS(Bid),
    ->     CONSTRAINT fk_noofcopies_ilibrary FOREIGN KEY (blid) REFERENCES Ilibrary(Lid)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> CREATE TABLE AUTHOR (
    ->     Aid INT PRIMARY KEY,
    ->     Aname VARCHAR(100) NOT NULL,
    ->     email VARCHAR(100) UNIQUE,
    ->     phoneno VARCHAR(15)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> CREATE TABLE PUBLISHER (
    ->     Pid INT PRIMARY KEY,
    ->     Pname VARCHAR(100) NOT NULL
    -> );
Query OK, 0 rows affected (0.03 sec)
```

Command Prompt - mysql -u

```
mysql>
mysql> CREATE TABLE Writes (
    ->     Bid INT,
    ->     Aid INT,
    ->     pid INT,
    ->     PRIMARY KEY (Bid, Aid, pid),
    ->     CONSTRAINT fk_writes_books FOREIGN KEY (Bid) REFERENCES BOOKS(Bid),
    ->     CONSTRAINT fk_writes_author FOREIGN KEY (Aid) REFERENCES AUTHOR(Aid),
    ->     CONSTRAINT fk_writes_publisher FOREIGN KEY (pid) REFERENCES PUBLISHER(Pid)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> CREATE TABLE SELLER (
    ->     Sid INT PRIMARY KEY,
    ->     slname VARCHAR(100) NOT NULL,
    ->     city VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> CREATE TABLE DEPARTMENT (
    ->     Deptid INT PRIMARY KEY,
    ->     deptname VARCHAR(100) NOT NULL,
    ->     Iname VARCHAR(100),
    ->     lid INT,
    ->     CONSTRAINT fk_dept_ilibrary FOREIGN KEY (lid) REFERENCES Ilibrary(Lid)
    -> );
Query OK, 0 rows affected (0.07 sec)

mysql>
mysql> CREATE TABLE Member (
    ->     memid INT PRIMARY KEY,
    ->     lid INT,
    ->     CONSTRAINT fk_member_ilibrary FOREIGN KEY (lid) REFERENCES Ilibrary(Lid)
    -> );
Query OK, 0 rows affected (0.08 sec)

mysql>
mysql> CREATE TABLE STUDENT (
    ->     Stuid INT PRIMARY KEY,
    ->     Sname VARCHAR(100) NOT NULL,
    ->     email VARCHAR(100) UNIQUE,
    ->     memid INT,
    ->     deptid INT,
    ->     CONSTRAINT fk_student_member FOREIGN KEY (memid) REFERENCES Member(memid),
    ->     CONSTRAINT fk_student_dept FOREIGN KEY (deptid) REFERENCES DEPARTMENT(Deptid)
    -> );
Query OK, 0 rows affected (0.10 sec)
```

Command Prompt - mysql -u

```
mysql>
mysql> CREATE TABLE STAFF (
    ->     Stid INT PRIMARY KEY,
    ->     stname VARCHAR(100) NOT NULL,
    ->     email VARCHAR(100) UNIQUE,
    ->     deptid INT,
    ->     memid INT,
    ->     CONSTRAINT fk_staff_dept FOREIGN KEY (deptid) REFERENCES DEPARTMENT(Deptid),
    ->     CONSTRAINT fk_staff_member FOREIGN KEY (memid) REFERENCES Member(memid)
    -> );
Query OK, 0 rows affected (0.08 sec)

mysql>
mysql> CREATE TABLE PURCHASE (
    ->     prid INT PRIMARY KEY,
    ->     lid INT,
    ->     sid INT,
    ->     pid INT,
    ->     bid INT,
    ->     quantity INT CHECK (quantity > 0),
    ->     date DATE,
    ->     totalcost DECIMAL(12, 2),
    ->     CONSTRAINT fk_purchase_ilibrary FOREIGN KEY (lid) REFERENCES Ilibrary(Lid),
    ->     CONSTRAINT fk_purchase_seller FOREIGN KEY (sid) REFERENCES SELLER(Sid),
    ->     CONSTRAINT fk_purchase_publisher FOREIGN KEY (pid) REFERENCES PUBLISHER(Pid),
    ->     CONSTRAINT fk_purchase_books FOREIGN KEY (bid) REFERENCES BOOKS(Bid)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> CREATE TABLE ISSUE (
    ->     lssueid INT PRIMARY KEY,
    ->     memid INT,
    ->     bid INT,
    ->     lid INT,
    ->     issuedate DATE,
    ->     returndate DATE,
    ->     CONSTRAINT fk_issue_member FOREIGN KEY (memid) REFERENCES Member(memid),
    ->     CONSTRAINT fk_issue_books FOREIGN KEY (bid) REFERENCES BOOKS(Bid),
    ->     CONSTRAINT fk_issue_ilibrary FOREIGN KEY (lid) REFERENCES Ilibrary(Lid)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> CREATE TABLE SELLS (
    ->     sid INT,
    ->     bid INT,
    ->     pid INT,
    ->     PRIMARY KEY (sid, bid, pid),
    ->     CONSTRAINT fk_sells_seller FOREIGN KEY (sid) REFERENCES SELLER(Sid),
    ->     CONSTRAINT fk_sells_books FOREIGN KEY (bid) REFERENCES BOOKS(Bid),
    ->     CONSTRAINT fk_sells_publisher FOREIGN KEY (pid) REFERENCES PUBLISHER(Pid)
    -> );
Query OK, 0 rows affected (0.05 sec)
```

```
mysql>
mysql> CREATE TABLE Employee (
    ->     eid INT PRIMARY KEY,
    ->     empname VARCHAR(100) NOT NULL,
    ->     email VARCHAR(100) UNIQUE,
    ->     salary DECIMAL(10, 2) CHECK (salary >= 0),
    ->     lid INT,
    ->     CONSTRAINT fk_emp_ilibrary FOREIGN KEY (lid) REFERENCES Ilibrary(Lid)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> CREATE TABLE A_specialization (
    ->     spec_id INT PRIMARY KEY,
    ->     spec_name VARCHAR(100) NOT NULL,
    ->     Aid INT,
    ->     CONSTRAINT fk_aspec_author FOREIGN KEY (Aid) REFERENCES AUTHOR(Aid) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.09 sec)

mysql>
```

```
mysql> show tables;
+---------------------+
| Tables_in_librarydb |
+---------------------+
| a_specialization    |
| author              |
| books               |
| department          |
| employee            |
| ilibrary            |
| issue               |
| member              |
| noofcopies          |
| publisher           |
| purchase            |
| seller              |
| sells               |
| siulibrary          |
| staff               |
| student             |
| writes              |
+---------------------+
17 rows in set (0.00 sec)

mysql>
```

Section-2(Q-2)

INSERT INTO SIULIBRARY (Slid, lname, location, noofbranches) VALUES

(1, 'SIU Pune Libraries', 'Pune', 3),

(2, 'SIU Bangalore Libraries', 'Bangalore', 2),

(3, 'SIU Nashik Libraries', 'Nashik', 1),

(4, 'SIU Noida Libraries', 'Noida', 1),

(5, 'SIU Hyderabad Libraries', 'Hyderabad', 1);

```sql
INSERT INTO ILIBRARY (Lid, lname, city, area, Slid) VALUES
(101, 'SIT Library', 'Pune', 'Lavale', 1),
(102, 'SIBM Library', 'Pune', 'Hinjawadi', 1),
(103, 'SICSR Library', 'Pune', 'Model Colony', 1),
(104, 'SSBS Library', 'Bangalore', NULL, 2),
(105, 'SCMS Library', 'Bangalore', 'Electronic City', 2);

INSERT INTO PUBLISHER (Pid, Pname) VALUES
(401, 'McGraw Hill'),
(402, 'Pearson'),
(403, 'Wiley Publications'),
(404, 'Technical Publications'),
(405, 'Oxford University Press');

INSERT INTO AUTHOR (Aid, Aname, email, phoneno) VALUES
(301, 'Shruti', 'shruti.author@example.com', '9876501001'),
(302, 'Shivam Kapoor', 'shivam.k@example.com', '9876501002'),
(303, 'Brian Kernighan', 'brian.k@example.com', '9876501003'),
(304, 'Ken Thompson', 'ken.t@example.com', '9876501004'),
(305, 'Ken Coel', 'ken.c@example.com', '9876501005');

INSERT INTO SELLER (Sid, slname, city) VALUES
(501, 'Pragati Book Store', 'Pune'),
(502, 'College Book Store', 'Bangalore'),
(503, 'Geeta', 'Mumbai'),
(504, 'Vidyarthi Book Depot', 'Nashik'),
(505, 'Pustak Seva', 'Pune');

INSERT INTO DEPARTMENT (Deptid, deptname, Iname, Lid) VALUES
(601, 'CS', 'SIT', 101),
(602, 'IT', 'SIT', 101),
(603, 'Civil', 'SIT', 101),
(604, 'E&TC', 'SIT', 101),
```

(605, 'Management', 'SIBM', 102);

INSERT INTO MEMBER (memid, Lid) VALUES
(701, 101),
(702, 101),
(703, 102),
(704, 104),
(705, 105);

INSERT INTO BOOKS (Bid, Bname, Price, Lid) VALUES
(201, 'Database Management Systems', 950.00, 101),
(202, 'Computer Networks', 750.00, 101),
(203, 'Data Structures Using C', 1200.00, 102),
(204, 'Advanced Artificial Intelligence', 12500.00, 104),
(205, 'Operating System Concepts', 850.00, 105);

INSERT INTO STUDENT (Stuid, Sname, email, memid, deptid) VALUES
(801, 'Rohit Patil', 'rohit.patil@student.siu.edu', 701, 603),
(802, 'Shivani More', 'shivani.more@student.siu.edu', 702, 601),
(803, 'Mayank Shah', 'mayank.shah@student.siu.edu', 703, 605),
(804, 'Priya Nair', 'priya.nair@student.siu.edu', 704, 604),
(805, 'Rohan Deshmukh', 'rohan.d@student.siu.edu', 705, 602);

INSERT INTO STAFF (Stid, stname, email, deptid, memid) VALUES
(901, 'Amit Joshi', 'amit.joshi@siu.edu', 601, 701),
(902, 'Sunil Patil', 'sunil.patil@siu.edu', 602, 702),
(903, 'Neha Sharma', 'neha.sharma@siu.edu', 603, 703),
(904, 'Anjali Rao', 'anjali.rao@siu.edu', 604, 704),
(905, 'Satish Kumar', 'satish.kumar@siu.edu', 605, 705);

INSERT INTO EMPLOYEE (eid, empname, email, salary, Lid) VALUES
(1001, 'Kiran Patil', 'kiran.patil@siu.edu', 45000.00, 101),
(1002, 'Meena Joshi', 'meena.joshi@siu.edu', 50000.00, 102),
(1003, 'Rahul Shah', 'rahul.shah@siu.edu', 62000.00, 103),

```sql
(1004, 'Pooja Nair', 'pooja.nair@siu.edu', 48000.00, 104),
(1005, 'Sameer Kulkarni', 'sameer.k@siu.edu', 70000.00, 105);

INSERT INTO A_SPECIALIZATION (spec_id, spec_name, Aid) VALUES
(1101, 'Database Systems', 301),
(1102, 'Software Engineering', 302),
(1103, 'C Programming', 303),
(1104, 'Operating Systems', 304),
(1105, 'Computer Networks', 305);

INSERT INTO WRITES (Bid, Aid, Pid) VALUES
(201, 301, 401),
(202, 305, 402),
(203, 303, 401),
(204, 302, 403),
(205, 304, 405);

INSERT INTO SELLS (Sid, Bid, Pid) VALUES
(501, 201, 401),
(501, 203, 401),
(502, 204, 403),
(503, 202, 402),
(504, 205, 405);

INSERT INTO NOOFCOPIES (bnid, bid, blid) VALUES
(1201, 201, 101),
(1202, 202, 101),
(1203, 203, 102),
(1204, 204, 104),
(1205, 205, 105);

INSERT INTO PURCHASE (prid, Lid, Sid, Pid, Bid, quantity, date, totalcost) VALUES
(1301, 101, 501, 401, 201, 8, '2026-01-15', 7600.00),
(1302, 101, 503, 402, 202, 5, '2026-03-10', 3750.00),
```

```
(1303, 102, 501, 401, 203, 10, '2026-06-18', 12000.00),
(1304, 104, 502, 403, 204, 4, '2026-07-05', 50000.00),
(1305, 105, 504, 405, 205, 7, '2026-12-12', 5950.00);

INSERT INTO ISSUE (lssueid, memid, Bid, Lid, issuedate, returndate) VALUES
(1401, 701, 201, 101, '2026-01-10', '2026-01-25'),
(1402, 702, 202, 101, '2026-02-12', '2026-02-27'),
(1403, 703, 203, 102, '2026-03-08', '2026-03-23'),
(1404, 704, 204, 104, '2026-04-15', '2026-04-30'),
(1405, 705, 205, 105, '2026-05-20', '2026-06-04');
```

Command Prompt - mysql -u

```
mysql> SELECT * FROM SIULIBRARY;
+------+-------------------------+-----------+--------------+
| Slid | lname                   | location  | noofbranches |
+------+-------------------------+-----------+--------------+
|    1 | SIU Pune Libraries      | Pune      |            3 |
|    2 | SIU Bangalore Libraries | Bangalore |            2 |
|    3 | SIU Nashik Libraries    | Nashik    |            1 |
|    4 | SIU Noida Libraries     | Noida     |            1 |
|    5 | SIU Hyderabad Libraries | Hyderabad |            1 |
+------+-------------------------+-----------+--------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM Ilibrary;
+-----+---------------+-----------+-----------------+------+
| Lid | lname         | city      | area            | slid |
+-----+---------------+-----------+-----------------+------+
| 101 | SIT Library   | Pune      | Lavale          |    1 |
| 102 | SIBM Library  | Pune      | Hinjawadi       |    1 |
| 103 | SICSR Library | Pune      | Model Colony    |    1 |
| 104 | SSBS Library  | Bangalore | NULL            |    2 |
| 105 | SCMS Library  | Bangalore | Electronic City |    2 |
+-----+---------------+-----------+-----------------+------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM BOOKS;
+-----+----------------------------------+----------+------+
| Bid | Bname                            | Price    | Lid  |
+-----+----------------------------------+----------+------+
| 201 | Database Management Systems      |   950.00 |  101 |
| 202 | Computer Networks                |   750.00 |  101 |
| 203 | Data Structures Using C          |  1200.00 |  102 |
| 204 | Advanced Artificial Intelligence | 12500.00 |  104 |
| 205 | Operating System Concepts        |   850.00 |  105 |
+-----+----------------------------------+----------+------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM Noofcopies;
+------+-----+------+
| bnid | bid | blid |
+------+-----+------+
| 1201 | 201 |  101 |
| 1202 | 202 |  101 |
| 1203 | 203 |  102 |
| 1204 | 204 |  104 |
| 1205 | 205 |  105 |
+------+-----+------+
5 rows in set (0.00 sec)
```

Command Prompt - mysql -u

```
mysql> SELECT * FROM AUTHOR;
+-----+-----------------+---------------------------+------------+
| Aid | Aname           | email                     | phoneno    |
+-----+-----------------+---------------------------+------------+
| 301 | Shruti          | shruti.author@example.com | 9876501001 |
| 302 | Shivam Kapoor   | shivam.k@example.com      | 9876501002 |
| 303 | Brian Kernighan | brian.k@example.com       | 9876501003 |
| 304 | Ken Thompson    | ken.t@example.com         | 9876501004 |
| 305 | Ken Coel        | ken.c@example.com         | 9876501005 |
+-----+-----------------+---------------------------+------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM PUBLISHER;
+-----+-------------------------+
| Pid | Pname                   |
+-----+-------------------------+
| 401 | McGraw Hill             |
| 402 | Pearson                 |
| 403 | Wiley Publications      |
| 404 | Technical Publications  |
| 405 | Oxford University Press |
+-----+-------------------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM Writes;
+-----+-----+-----+
| Bid | Aid | pid |
+-----+-----+-----+
| 201 | 301 | 401 |
| 204 | 302 | 403 |
| 203 | 303 | 401 |
| 205 | 304 | 405 |
| 202 | 305 | 402 |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT * FROM SELLER;
+-----+----------------------+-----------+
| Sid | slname               | city      |
+-----+----------------------+-----------+
| 501 | Pragati Book Store   | Pune      |
| 502 | College Book Store   | Bangalore |
| 503 | Geeta                | Mumbai    |
| 504 | Vidyarthi Book Depot | Nashik    |
| 505 | Pustak Seva          | Pune      |
+-----+----------------------+-----------+
5 rows in set (0.00 sec)
```

Command Prompt - mysql -u

```
mysql> SELECT * FROM DEPARTMENT;
+--------+------------+-------+------+
| Deptid | deptname   | Iname | lid  |
+--------+------------+-------+------+
|    601 | CS         | SIT   |  101 |
|    602 | IT         | SIT   |  101 |
|    603 | Civil      | SIT   |  101 |
|    604 | E&TC       | SIT   |  101 |
|    605 | Management | SIBM  |  102 |
+--------+------------+-------+------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM Member;
+-------+------+
| memid | lid  |
+-------+------+
|   701 |  101 |
|   702 |  101 |
|   703 |  102 |
|   704 |  104 |
|   705 |  105 |
+-------+------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM STUDENT;
+-------+----------------+-------------------------------+-------+--------+
| Stuid | Sname          | email                         | memid | deptid |
+-------+----------------+-------------------------------+-------+--------+
|   801 | Rohit Patil    | rohit.patil@student.siu.edu   |   701 |    603 |
|   802 | Shivani More   | shivani.more@student.siu.edu  |   702 |    601 |
|   803 | Mayank Shah    | mayank.shah@student.siu.edu   |   703 |    605 |
|   804 | Priya Nair     | priya.nair@student.siu.edu    |   704 |    604 |
|   805 | Rohan Deshmukh | rohan.d@student.siu.edu       |   705 |    602 |
+-------+----------------+-------------------------------+-------+--------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM STAFF;
+------+--------------+-----------------------+--------+-------+
| Stid | stname       | email                 | deptid | memid |
+------+--------------+-----------------------+--------+-------+
|  901 | Amit Joshi   | amit.joshi@siu.edu    |    601 |   701 |
|  902 | Sunil Patil  | sunil.patil@siu.edu   |    602 |   702 |
|  903 | Neha Sharma  | neha.sharma@siu.edu   |    603 |   703 |
|  904 | Anjali Rao   | anjali.rao@siu.edu    |    604 |   704 |
|  905 | Satish Kumar | satish.kumar@siu.edu  |    605 |   705 |
+------+--------------+-----------------------+--------+-------+
5 rows in set (0.00 sec)
```

Command Prompt - mysql -u

```
mysql> SELECT * FROM PURCHASE;
+------+------+------+------+------+----------+------------+-----------+
| prid | lid  | sid  | pid  | bid  | quantity | date       | totalcost |
+------+------+------+------+------+----------+------------+-----------+
| 1301 |  101 |  501 |  401 |  201 |        8 | 2026-01-15 |   7600.00 |
| 1302 |  101 |  503 |  402 |  202 |        5 | 2026-03-10 |   3750.00 |
| 1303 |  102 |  501 |  401 |  203 |       10 | 2026-06-18 |  12000.00 |
| 1304 |  104 |  502 |  403 |  204 |        4 | 2026-07-05 |  50000.00 |
| 1305 |  105 |  504 |  405 |  205 |        7 | 2026-12-12 |   5950.00 |
+------+------+------+------+------+----------+------------+-----------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM ISSUE;
+---------+-------+------+------+------------+------------+
| lssueid | memid | bid  | lid  | issuedate  | returndate |
+---------+-------+------+------+------------+------------+
|    1401 |   701 |  201 |  101 | 2026-01-10 | 2026-01-25 |
|    1402 |   702 |  202 |  101 | 2026-02-12 | 2026-02-27 |
|    1403 |   703 |  203 |  102 | 2026-03-08 | 2026-03-23 |
|    1404 |   704 |  204 |  104 | 2026-04-15 | 2026-04-30 |
|    1405 |   705 |  205 |  105 | 2026-05-20 | 2026-06-04 |
+---------+-------+------+------+------------+------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM SELLS;
+-----+-----+-----+
| sid | bid | pid |
+-----+-----+-----+
| 501 | 201 | 401 |
| 503 | 202 | 402 |
| 501 | 203 | 401 |
| 502 | 204 | 403 |
| 504 | 205 | 405 |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT * FROM Employee;
+------+-----------------+----------------------+----------+-----+
| eid  | empname         | email                | salary   | lid |
+------+-----------------+----------------------+----------+-----+
| 1001 | Kiran Patil     | kiran.patil@siu.edu  | 45000.00 | 101 |
| 1002 | Meena Joshi     | meena.joshi@siu.edu  | 50000.00 | 102 |
| 1003 | Rahul Shah      | rahul.shah@siu.edu   | 62000.00 | 103 |
| 1004 | Pooja Nair      | pooja.nair@siu.edu   | 48000.00 | 104 |
| 1005 | Sameer Kulkarni | sameer.k@siu.edu     | 70000.00 | 105 |
+------+-----------------+----------------------+----------+-----+
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM A_specialization;
+---------+----------------------+------+
| spec_id | spec_name            | Aid  |
+---------+----------------------+------+
|    1101 | Database Systems     |  301 |
|    1102 | Software Engineering |  302 |
|    1103 | C Programming        |  303 |
|    1104 | Operating Systems    |  304 |
|    1105 | Computer Networks    |  305 |
+---------+----------------------+------+
5 rows in set (0.00 sec)
```

Section-2(Q-3)

SELECT Lid, lname, city, area

FROM Ilibrary

WHERE LOWER(city) = 'pune';

SELECT deptname, Iname

FROM DEPARTMENT

WHERE UPPER(deptname) = 'CS';

SELECT Bid, Bname, Price

FROM BOOKS

WHERE Price BETWEEN 800 AND 12000;

SELECT eid, empname, salary

FROM Employee

WHERE salary <= 50000;

SELECT Sid, slname, city

FROM SELLER

WHERE LOWER(slname) LIKE '%ta';

SELECT Lid, lname, city, area

FROM Ilibrary

WHERE area IS NULL;

SELECT Stid, stname, email

```sql
FROM STAFF
WHERE UPPER(stname) NOT LIKE 'A%';

SELECT DISTINCT s.Slid, s.lname, s.location
FROM SIULIBRARY s
JOIN Ilibrary i ON s.Slid = i.slid
WHERE LOWER(i.city) = 'bangalore';

SELECT st.Stuid, st.Sname, d.deptname
FROM STUDENT st
JOIN DEPARTMENT d ON st.deptid = d.Deptid
WHERE LOWER(d.deptname) = 'civil';

SELECT b.Bid, b.Bname, a.Aname, p.Pname
FROM BOOKS b
JOIN Writes w ON b.Bid = w.Bid
JOIN AUTHOR a ON w.Aid = a.Aid
JOIN PUBLISHER p ON w.pid = p.Pid
WHERE LOWER(a.Aname) = 'shruti'
  AND LOWER(p.Pname) = 'mcgraw hill';
```

```
Command Prompt - mysql  -u

mysql> SELECT Lid, lname, city, area
    -> FROM Ilibrary
    -> WHERE LOWER(city) = 'pune';
+-----+---------------+------+--------------+
| Lid | lname         | city | area         |
+-----+---------------+------+--------------+
| 101 | SIT Library   | Pune | Lavale       |
| 102 | SIBM Library  | Pune | Hinjawadi    |
| 103 | SICSR Library | Pune | Model Colony |
+-----+---------------+------+--------------+
3 rows in set (0.01 sec)

mysql>
mysql> SELECT deptname, Iname
    -> FROM DEPARTMENT
    -> WHERE UPPER(deptname) = 'CS';
+----------+-------+
| deptname | Iname |
+----------+-------+
| CS       | SIT   |
+----------+-------+
1 row in set (0.00 sec)

mysql>
mysql> SELECT Bid, Bname, Price
    -> FROM BOOKS
    -> WHERE Price BETWEEN 800 AND 12000;
+-----+-----------------------------+---------+
| Bid | Bname                       | Price   |
+-----+-----------------------------+---------+
| 201 | Database Management Systems |  950.00 |
| 203 | Data Structures Using C     | 1200.00 |
| 205 | Operating System Concepts   |  850.00 |
+-----+-----------------------------+---------+
3 rows in set (0.00 sec)

mysql>
mysql> SELECT eid, empname, salary
    -> FROM Employee
    -> WHERE salary <= 50000;
+------+-------------+----------+
| eid  | empname     | salary   |
+------+-------------+----------+
| 1001 | Kiran Patil | 45000.00 |
| 1002 | Meena Joshi | 50000.00 |
| 1004 | Pooja Nair  | 48000.00 |
+------+-------------+----------+
3 rows in set (0.00 sec)
```

Command Prompt - mysql -u

```
mysql>
mysql> SELECT Sid, slname, city
    -> FROM SELLER
    -> WHERE LOWER(slname) LIKE '%ta';
+-----+--------+--------+
| Sid | slname | city   |
+-----+--------+--------+
| 503 | Geeta  | Mumbai |
+-----+--------+--------+
1 row in set (0.00 sec)

mysql>
mysql> SELECT Lid, lname, city, area
    -> FROM Ilibrary
    -> WHERE area IS NULL;
+-----+--------------+-----------+------+
| Lid | lname        | city      | area |
+-----+--------------+-----------+------+
| 104 | SSBS Library | Bangalore | NULL |
+-----+--------------+-----------+------+
1 row in set (0.00 sec)

mysql>
mysql> SELECT Stid, stname, email
    -> FROM STAFF
    -> WHERE UPPER(stname) NOT LIKE 'A%';
+------+--------------+----------------------+
| Stid | stname       | email                |
+------+--------------+----------------------+
|  902 | Sunil Patil  | sunil.patil@siu.edu  |
|  903 | Neha Sharma  | neha.sharma@siu.edu  |
|  905 | Satish Kumar | satish.kumar@siu.edu |
+------+--------------+----------------------+
3 rows in set (0.00 sec)

mysql>
mysql> SELECT DISTINCT s.Slid, s.lname, s.location
    -> FROM SIULIBRARY s
    -> JOIN Ilibrary i ON s.Slid = i.slid
    -> WHERE LOWER(i.city) = 'bangalore';
+------+--------------------------+-----------+
| Slid | lname                    | location  |
+------+--------------------------+-----------+
|    2 | SIU Bangalore Libraries  | Bangalore |
+------+--------------------------+-----------+
1 row in set (0.00 sec)
```

```
mysql>
mysql> SELECT st.Stuid, st.Sname, d.deptname
    -> FROM STUDENT st
    -> JOIN DEPARTMENT d ON st.deptid = d.Deptid
    -> WHERE LOWER(d.deptname) = 'civil';
+-------+-------------+----------+
| Stuid | Sname       | deptname |
+-------+-------------+----------+
|   801 | Rohit Patil | Civil    |
+-------+-------------+----------+
1 row in set (0.00 sec)

mysql>
mysql> SELECT b.Bid, b.Bname, a.Aname, p.Pname
    -> FROM BOOKS b
    -> JOIN Writes w ON b.Bid = w.Bid
    -> JOIN AUTHOR a ON w.Aid = a.Aid
    -> JOIN PUBLISHER p ON w.pid = p.Pid
    -> WHERE LOWER(a.Aname) = 'shruti'
    ->   AND LOWER(p.Pname) = 'mcgraw hill';
+-----+-----------------------------+--------+-------------+
| Bid | Bname                       | Aname  | Pname       |
+-----+-----------------------------+--------+-------------+
| 201 | Database Management Systems | Shruti | McGraw Hill |
+-----+-----------------------------+--------+-------------+
1 row in set (0.00 sec)

mysql>
```

Conclusion: All DDL and DML operations—including table creation with primary/foreign key constraints, sample data insertion, and data retrieval queries—were successfully written and executed to manage the Library Management System database efficiently.